Math Methods (PH3103)

HW-6

(1) $$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega}{2\hbar}x^2\right)$$

$$= a e^{-bx^2} \quad ;$$

$$a = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \quad ; \quad b = \frac{m\omega}{2\hbar}$$

$$\therefore \tilde{\psi}_0(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} \psi_0(x)\, dx.$$

$$= \frac{a}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-bx^2 - ikx}\, dx.$$

$$-bx^2 - ikx.$$

$$= -b\left(x^2 + \frac{ikx}{2b}\cdot 2 + \left(\frac{ik}{2b}\right)^2\right) + b\cdot\left(\frac{ik}{2b}\right)^2$$

$$= -b\left(x + \frac{ik}{2b}\right)^2 - \frac{k^2}{4b}.$$

$$\therefore \tilde{\psi}_0(k) = \frac{a}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-k^2/4b} \cdot e^{-b\left(x + \frac{ik}{2b}\right)^2} dx$$

$$= \frac{a e^{-k^2/4b}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-bz^2} dz \qquad \left[z = x + \frac{ik}{2b} \Rightarrow dz = dx \right].$$

$$= \frac{a e^{-k^2/4b}}{\sqrt{2\pi}} \cdot \sqrt{\frac{\pi}{b}} \qquad \left[\int_{-\infty}^{\infty} e^{-az^2} dz = \sqrt{\frac{\pi}{a}} \right]$$

$$= \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \cdot \frac{1}{\sqrt{2}} \cdot \sqrt{\frac{2\hbar}{m\omega}} \cdot \exp\left(-\frac{\hbar k^2}{2m\omega}\right).$$

$$\boxed{\tilde{\psi}_0(k) = \left(\frac{\hbar}{\pi m\omega}\right)^{1/4} \exp\left(-\frac{\hbar k^2}{2m\omega}\right)} \quad \text{(Ans)}$$

$$\langle x \rangle = \int_{-\infty}^{\infty} \psi_0^*(x)\, x\, \psi_0(x)\, dx = \sqrt{\frac{m\omega}{\pi\hbar}} \int_{-\infty}^{\infty} e^{-\frac{m\omega}{\hbar}x^2}\, x\, dx = 0$$

$$\boxed{\langle x \rangle = 0}$$

[odd integral]

$$\langle x^2 \rangle = a^2 \int_{-\infty}^{\infty} e^{-2b.x^2} \cdot x^2 \, dx$$

$$I(a) = \int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\pi}\, a^{-1/2}.$$

$$\Rightarrow \frac{dI(a)}{da} = +\int_{-\infty}^{\infty} x^2 e^{-ax^2} dx = +\frac{1}{2}\sqrt{\pi}\, a^{-3/2}$$

$$\Rightarrow \boxed{\int_{-\infty}^{\infty} x^2 e^{-ax^2} dx = \sqrt{\frac{\pi}{2}}\, a^{-3/2}}$$

$$\therefore \langle x^2 \rangle = a^2 \cdot \sqrt{\frac{\pi}{2}}\,(2b)^{-3/2}$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}} \cdot \sqrt{\frac{\pi}{2}} \cdot \left(\frac{m\omega}{\hbar}\right)^{-3/2}$$

$$= \sqrt{\frac{m\omega}{\hbar}} \cdot \frac{1}{2} \cdot \frac{\hbar}{m\omega}\sqrt{\frac{\hbar}{m\omega}} = \boxed{\frac{\hbar}{2m\omega} = \langle x^2 \rangle}$$

$$\therefore \langle p \rangle = -i\hbar a^2 \int_{-\infty}^{\infty} e^{-bx^2} \frac{d}{dx}\left(e^{-bx^2}\right) dx.$$

$$= +i\hbar a^2 b \int_{-\infty}^{\infty} e^{-2bx^2} \cdot 2x \, dx = 0 \quad [\text{odd integral}]$$

$$\boxed{\langle p \rangle = 0}.$$

$$\therefore \langle p^2 \rangle = -a^2\hbar^2 \int_{-\infty}^{\infty} e^{-bx^2} \frac{d^2}{dx^2}\left(e^{-bx^2}\right) dx.$$

$$= +2a^2\hbar^2 b \int e^{-bx^2} \frac{d}{dx}\left(e^{-bx^2} x\right) dx$$

$$= 2a^2\hbar^2 b \int e^{-2bx^2}\left(1 - 2bx^2\right) dx.$$

$$= 2a^2\hbar^2 b \left[\int e^{-2bx^2} dx - 2b\int e^{-2bx^2} x^2 dx\right].$$

$$= 2a^2\hbar^2 b \left[\sqrt{\frac{\pi}{2b}} - 2b \cdot \sqrt{\frac{\pi}{2}} \cdot (2b)^{-3/2}\right]$$

$$= 2a^2\hbar^2 b \left[\sqrt{\frac{\pi}{2b}} - \frac{1}{2}\sqrt{\frac{\pi}{2b}}\right].$$

$$= a^2\hbar^2 b \cdot \sqrt{\frac{\pi}{2b}}$$

$$= \sqrt{\frac{m\omega}{\pi\hbar}} \cdot \hbar^2 \cdot \frac{m\omega}{2\hbar} \cdot \sqrt{\frac{\pi\hbar}{m\omega}} = \boxed{\frac{m\omega\hbar}{2} = \langle p^2 \rangle}$$

$$\therefore \Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{\hbar}{2m\omega}}$$

$$\therefore \Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{\frac{m\omega\hbar}{2}}$$

$$\Delta x \Delta p = \frac{\hbar}{2} \quad \text{(Ans)}$$

② $y'' - \mu^2 y = \delta(x - x_0)$

$$F(y''(x)) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \underbrace{y''(x)}_{\text{(ii)}} \underbrace{e^{-ikx}}_{\text{(i)}} dx$$

$$= \frac{1}{\sqrt{2\pi}} \left[\cancel{e^{-ikx} y'(x) \Big|_{-\infty}^{\infty}}^{\,0} + ik \int_{-\infty}^{\infty} e^{-ikx} y'(x)\, dx \right]$$

$$= \frac{ik}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \underbrace{e^{-ikx}}_{\text{(i)}} \underbrace{y'(x)}_{\text{(ii)}} dx$$

[$y'(x)' = 0 = y(x)$ @ $x \to \pm\infty$]

$$= \frac{ik}{\sqrt{2\pi}} \left[\cancel{e^{-ikx} y(x)}^{\,0} + ik \int_{-\infty}^{\infty} e^{-ikx} y(x)\, dx \right]$$

$$= -k^2 \tilde{y}(k) \quad \text{— (i)}$$

$$\therefore F(y(x)) = \tilde{y}(k) \quad \text{— (ii)}$$

$$\therefore F[\delta(x - x_0)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x - x_0) e^{-ikx} dx = \frac{e^{-ikx_0}}{\sqrt{2\pi}} \quad \text{— (iii)}$$

Plugging (i), (ii) & (iii) into the ODE:

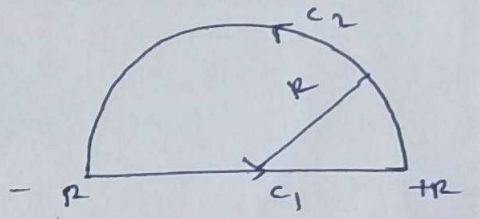
$$-k^2 \tilde{y}(k) - \mu^2 \tilde{y}(k) = e^{-ikx_0}/\sqrt{2\pi}$$

[we Fourier-transformed every term of the ODE]

$$\Rightarrow \boxed{\tilde{y}(k) = -\frac{e^{-ikx_0}}{\sqrt{2\pi}\,(k^2 + \mu^2)}}$$

$$\therefore y(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{y}(k) e^{ikx} dk$$

$$= -\frac{1}{2\pi} \underbrace{\boxed{\int_{-\infty}^{\infty} \frac{e^{+ik(x - x_0)}}{k^2 + \mu^2} dk}}_{\to I}$$



Let $k = z$; $x - x_0 = a$

$$\therefore \int_{-\infty}^{\infty} \frac{e^{iaz}}{z^2 + \mu^2} dz = \oint_{c} \frac{e^{iaz}}{z^2 + \mu^2} dz - \int_{c_2} \frac{e^{iaz}}{z^2 + \mu^2} dz \quad [c = c_1 + c_2]$$

$\oint_{c_2} \frac{e^{iaz}}{}$ Case 1: $x > x_0 \to a > 0$.

we close c_2 in the upper-half of $\mathbb{C}$.

$$\int_{c_2} \underbrace{\frac{e^{iaz}}{z^2 + \mu^2}}_{f(z)} dz = 0 \quad [\text{Jordan's lemma}]$$

$\oint_{c} f(z)$ has a pole of order 1 @ $z_0 = i\mu$ as we are working in upper $\mathbb{C}$.

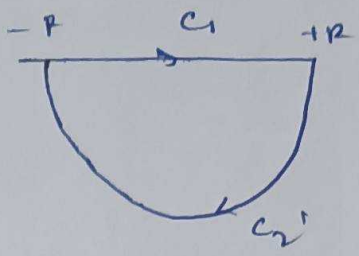
$$\therefore \oint_c f(z)\,dz = 2\pi i \operatorname{Res} f(z)\Big|_{z=i\mu}$$

$$= 2\pi i \cdot \frac{e^{iaz}}{z+i\mu}\Big|_{z=i\mu}$$

$$= \frac{\pi}{\mu} e^{-\mu a} = \boxed{\frac{\pi}{\mu} e^{-\mu(x-x_0)} = I}$$

<u>Case II</u>: $x < x_0 \Rightarrow a < 0$

We close C_2' in the lower-half of ϕ.



$$\therefore \int_{C_2'} f(z)\,dz = 0 \quad [\text{Jordan's Lemma}]$$

$C' = C_1 + C_2'$

$\therefore$ $f(z)$ has a pole @ $z_0 = -i\mu$ as we are working in lower ϕ.

$$\therefore \oint_{C'} f(z)\,dz = -2\pi i \frac{e^{iaz}}{z-i\mu}\Big|_{z=-i\mu} \quad [\text{Clockwise direction}]$$

$$= \frac{\pi}{\mu} e^{\mu a} = \boxed{\frac{\pi}{\mu} e^{\mu(x-x_0)} = I}$$

Combining the 2 cases:

$$I = \frac{\pi}{\mu} e^{-\mu|x-x_0|}$$

$$\Rightarrow \boxed{y(x) = -\frac{1}{2\mu} e^{-\mu|x-x_0|}} \quad (\text{Ans})$$

<u>Verificat^n</u>: $[x \geq x_0]$

$$y'' - \mu^2 y = 0.$$

$$\therefore y'(x) = \frac{1}{2} e^{-\mu(x-x_0)} \quad ; \quad y''(x) = -\frac{\mu}{2} e^{-\mu(x-x_0)}$$

$$\therefore y'' - \mu^2 y = -\frac{\mu}{2} e^{-\mu(x-x_0)} + \frac{\mu}{2} e^{-\mu(x-x_0)} = 0$$

[satisfies the ODE]

<u>$x < x_0$</u>: $y'' - \mu^2 y = 0$

$$y(x) = -\frac{1}{2\mu} e^{\mu(x-x_0)} \Rightarrow y'(x) = -\frac{1}{2} e^{\mu(x-x_0)}$$

$$\Rightarrow y''(x) = -\frac{\mu}{2} e^{\mu(x-x_0)}$$

$\therefore\ y'' - \mu^2 y = -\frac{\mu}{2} e^{\mu(x-x_0)} + \frac{\mu}{2} e^{\mu(x-x_0)} = 0$ [verified]

(10)

(3) $y'' - \mu^2 y = \cos bx.$

$\therefore\ F(y'') = -k^2 \tilde{y}(k)$ — (i)

$F(y) = \tilde{y}(k)$ — (ii)

$$F(\cos bx) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} \cos bx \, dx.$$

$$= \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} \left[e^{ibx} + e^{-ibx} \right] dx$$

$$= \frac{1}{2\sqrt{2}} \left[\int e^{-i(k-b)x} dx + \int e^{-i(k+b)x} dx \right]$$

$$= \frac{1}{2\sqrt{2\pi}} \left[2\pi \delta(k-b) + 2\pi \delta(k+b) \right]$$

$$= \frac{\pi}{\sqrt{2\pi}} \left[\delta(k-b) + \delta(k+b) \right] \quad \left[\int_{-\infty}^{\infty} e^{i(k-b)x} dx = 2\pi\delta(k-b) \right.$$

— (iii) $\left. \delta(k-b) = \delta(b-k) \right]$

Plugging (i), (ii) & (iii) into the ODE:

$$(-k^2 - \mu^2)\, \tilde{y}(k) = \frac{\pi}{\sqrt{2}} \left[\delta(k-b) + \delta(k+b) \right] \quad \text{[RT of ODE]}$$

$$\Rightarrow \boxed{\tilde{y}(k) = -\sqrt{\frac{\pi}{2}} \frac{\left[\delta(k-b) + \delta(k+b) \right]}{(k^2 + \mu^2)}}$$

$$\therefore\ y(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{y}(k)\, e^{ikx} dk$$

$$= -\frac{1}{2} \left[\int_{-\infty}^{\infty} \frac{\delta(k-b) e^{ikx}}{k^2 + \mu^2} dk + \int_{-\infty}^{\infty} \frac{\delta(k+b)\, e^{ikx}}{k^2 + \mu^2} dk \right]$$

$$= -\frac{1}{2} \left[\frac{e^{ibx}}{b^2 + \mu^2} + \frac{e^{-ibx}}{k^2 + \mu^2} \right]$$

$$\boxed{y(x) = -\frac{\cos bx}{b^2 + \mu^2}}$$ (Ans)